Date: / /

* Find the first & Follow for the following grammar.

$S \rightarrow ABCDE$

$A \rightarrow a / \epsilon$

$B \rightarrow b / \epsilon$

$C \rightarrow c$

$D \rightarrow d / \epsilon$

$E \rightarrow e / \epsilon$

$\rightarrow$

FIRST (S) = FIRST (A B C D E)

= FIRST (A) = {a, FIRST(B)} = {a, b, c}

FIRST (A) = {a, ε}

FIRST (B) = {b, ε}

FIRST (C) = {c}

FIRST (D) = {d, ε}

FIRST (E) = {e, ε}

FOLLOW (S) = {$}

FOLLOW (A) = { FIRST (B)}

= {b, c}

FOLLOW(B) = { FIRST (C)} = {c}

FOLLOW (C) = { FIRST(D)} = {d, FOLLOW(D)}

= {d, FIRST (E)}

= {d, e, Follow (S)}

= {d, e, $}

FOLLOW (D) = {FIRST (E)} = {e, Follow(S)}

= {ε, $}

FOLLOW(E) = {follow(S) = {$}

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LL(1) Parser ~~tree~~ table

	a	b	c	d	e	$
S	S→ABCDE	S→ABCDE	S→ABCDE			
A	A→a	A→ϵ	A→ϵ			
B		B→b	B→ϵ			
C			C→c			
D				D→d	D→ϵ	D→ϵ
E					E→e	E→ϵ

There are all single Entities in each ~~the~~ column & row
∴ it is LL(1)

Q2. Find out First & Follow of the following grammar & check wheather it is LL(1) or not.

1) S → Bb | Cd
B → aB | ϵ
C → cC | ϵ

→ FIRST(S) = {a, c, b, d}
FIRST(B) = {a, ϵ}
FIRST(C) = {c, ϵ}

FOLLOW(S) = {$}
FOLLOW(B) = {b, ~~FIRST(B)~~ FOLLOW(B)} = {b}
FOLLOW(C) = {d, ~~FOLLOW(C)~~} = {d}

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	a	b	c	d	$
S	$S \to Bb$	$S \to Bb$	$S \to cd$	$S \to cd$	
B	$B \to aB$	$B \to \epsilon$			
C			$C \to cC$	$C \to \epsilon$	